Invenergy

Invenergy is a renewable energy company that develops utility-scale projects. Before construction can begin, a project must get a permit from local state boards, who review public comments from community members to make their decision. Because manually reading and summarizing hundreds of comments was slow, Invenergy's team struggled to see patterns and respond to concerns in real time, increasing the risk of lost investments due to permit rejection.

Invenergy asked UChicago's Data Science Clinic to develop an automated text processing pipeline that summarizes the sentiment and main themes present in public comments submitted for projects in development. The team built an end-to-end pipeline capable of scraping comments from public utility websites across nine states and extracting text from a range of document types, including typed and handwritten text. Next, the team developed deep learning models to tag comments as supportive, opposed or neutral, and identify the main issues raised (e.g. flooding, loss of farmland, etc.) These results are visualized with an interactive dashboard (Figure 1) that gives Invenergy a fast and consistent way to understand community sentiment towards a project or compare across projects. The team also reproduced this pipeline in Invenergy's Databricks environment to ensure the pipeline's full reusability.

Figure 1: *Subset of dashboard statistics generated for an active project (with U-2196 as the docket ID for the Sloopy Solar Project), showing both the distribution of sentiment and primary areas of concern across submitted comments.*